

PROPOSAL

Development of a Web-Based Nursing Staffing Tool Using WHO WISN Methodology for Nepalese Hospitals

Submitted to

Department of Health Informatics

Submitted by

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Submission Date : 03/10/2026

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Chapter 1: Introduction

Human resources are widely recognized as the most critical asset in any functioning health system, and nurses represent the backbone of healthcare delivery in hospitals worldwide. In Nepal, as in many low- and middle-income countries, nursing staff perform an expansive range of responsibilities: continuous patient monitoring, medication administration, comprehensive documentation, care coordination across departments, infection control, and direct patient support. Despite the central role that nurses play, nursing workforce management in Nepalese hospitals continues to face profound challenges rooted in persistent staff shortages, uneven geographical distribution, rapidly increasing service demands, and the absence of evidence-based systems for workforce planning.

The consequences of inadequate nursing workforce planning extend well beyond administrative inconvenience. When hospitals rely on intuition, fixed historical norms, or informal observation to determine staffing levels, some departments become critically overloaded while others remain underutilized. This imbalance drives nurse burnout, increases medical errors, compromises patient safety, and ultimately weakens health outcomes. Research conducted across multiple countries confirms that inappropriate health workforce planning interventions contribute significantly to critical workforce shortages and weakened health systems (Al-Sawai & Al-Shishtawy, 2015). In Nepal specifically, these challenges are compounded by a severe urban-rural divide in nurse distribution, leaving rural health facilities with dangerously insufficient staffing while urban hospitals remain better resourced but still inadequately managed.

Recognizing these global challenges, the World Health Organization (WHO) developed the Workload Indicators of Staffing Need (WISN) methodology as a systematic, evidence-based approach to calculating staffing requirements. WISN evaluates three core components: the available working time of health workers, the

actual workload they are expected to perform, and the professionally established standards for how long each service activity should take. By combining these three elements, WISN produces an objective calculation of the number of staff required in any given health facility or department, as well as a WISN ratio that indicates whether current staffing levels are adequate, insufficient, or in surplus (WHO, 2010, 2023).

WISN methodology has been officially recommended for workforce planning in Nepal and has been successfully applied in countries across Africa, Asia, and South America. In South Africa, WISN helped identify that apparent nurse shortages were partly driven by nurses performing non-clinical tasks such as administrative work and facility cleaning due to inadequate support staff (Ravhengani & Mtshali, 2017). In Germany, a WISN application in a hospital neurology department not only calculated precise staffing needs but identified process inefficiencies that, once addressed, could effectively free the equivalent of 1.36 full-time therapists for direct patient care (Thum, Wehner & Goetz, 2024). These examples illustrate WISN's power not merely as a counting tool but as a management and optimization instrument.

Despite its scientific validity and global recognition, practical adoption of WISN in Nepalese hospitals remains severely limited. The primary obstacle is implementation complexity. Most WISN applications rely on manual calculations performed using Microsoft Excel spreadsheets, a process that is labor-intensive, prone to human error, difficult to update regularly, and not designed for routine managerial use. Hospital administrators and nurse managers frequently find Excel-based WISN tools technically demanding, resulting in WISN being applied as a one-time analytical exercise rather than an integrated component of ongoing workforce management. This gap between the theoretical power of WISN and its practical usability represents the central challenge this project seeks to address.

The rapid development of digital health technologies and health informatics now presents a transformative opportunity to bridge this gap. A well-designed web-based application can automate all WISN calculations, eliminate manual errors, provide

instant results, and present complex staffing information through intuitive visual dashboards that enable confident decision-making by managers with limited technical backgrounds. By translating WISN's established formulas into a user-friendly digital tool, this project aims to convert a powerful but underutilized methodology into a practical, accessible instrument for routine hospital management.

This proposal describes the development of a web-based nursing staffing tool that implements WHO WISN methodology for Nepalese hospital settings. The project combines established public health science with modern health informatics principles to create a prototype system that hospital administrators, nurse managers, and health policymakers can use to make evidence-based staffing decisions. The tool will be designed with the specific context of resource-constrained Nepalese hospitals in mind, ensuring accessibility through standard internet browsers, compatibility with limited technical infrastructure, and an interface designed for users with varying levels of digital literacy.

The following proposal outlines the background and problem context, states the project objectives, reviews the relevant literature, describes the methodology for system development and validation using simulated data, presents expected outcomes, provides a project timeline, discusses the project's significance and limitations, and offers a concluding summary. Together, these sections demonstrate the academic rigor, practical relevance, and feasibility of developing a digital WISN tool for Nepalese hospitals.

Chapter 2: Background and Problem Statement

Nepal's health system operates under significant resource constraints, with nursing workforce challenges representing one of the most pressing areas of concern. The country faces a complex combination of absolute shortage in the total number of trained nurses, severe maldistribution between urban and rural areas, and inadequate management systems that prevent efficient utilization of the nurses who are available. According to Nepal's Ministry of Health and Population (2012), the nursing workforce is disproportionately concentrated in urban centers, particularly in the Kathmandu Valley, while rural and remote districts often operate with a fraction of the staffing levels required to deliver basic health services.

The consequences of this workforce crisis manifest across every level of healthcare delivery. In hospital settings, nurses who are present are frequently overworked, managing patient loads that far exceed safe ratios. Research on nurses' workload and patient safety in Nepalese hospitals demonstrates that excessive workloads directly compromise care quality, increase medication errors, and contribute to high rates of nurse burnout and turnover (Shrestha & Kadel, 2019). Paradoxically, poorly designed workforce planning systems mean that even in facilities with adequate nurse numbers, poor distribution of staff across departments creates pockets of extreme overload alongside departments with unnecessary surplus.

A particularly important insight from international research is that nursing shortages are not always what they appear. A qualitative study of primary health care nursing management in South Africa found that WISN analysis revealed no actual nursing shortage in absolute terms; rather, professional nurses were spending disproportionate time on non-clinical activities such as administrative documentation, facility cleaning, and support tasks that should be performed by lower-skilled auxiliary staff (Ravhengani & Mtshali, 2017). This finding has direct relevance to Nepal, where similar patterns of

task shifting from support staff to professional nurses are commonly observed, effectively reducing nursing capacity for clinical care.

The WHO WISN methodology was specifically designed to provide an objective, evidence-based response to these types of workforce planning challenges. First developed in the 1990s and substantially updated in 2010 and again in 2023, WISN provides a systematic eight-step process for calculating the number of health workers of a given cadre needed to perform all the health service activities at a given health facility or department (WHO, 2010, 2023). Unlike population-to-staff ratios or fixed establishment norms, WISN grounds staffing calculations in actual workload data, making its recommendations directly responsive to the specific service volumes and activity mix of any given facility.

Despite WISN's widespread international adoption and Nepal's formal endorsement of the methodology, actual implementation at the facility level remains rare and sporadic. The barriers to implementation are well-documented. A multi-country analysis of WISN implementation experiences in India, South Africa, and Peru found that common barriers included inadequate stakeholder consultation, information system gaps that made it difficult to obtain reliable workload data, insufficient technical support for facility managers, and the substantial time burden required for manual WISN calculations (Mabunda et al., 2021). In South Africa's case, even after successful WISN implementation at the national level, the methodology was ultimately abandoned because the calculated staffing requirements were deemed unaffordable, highlighting the additional challenge that WISN must be paired with realistic resource planning to achieve lasting impact.

The introduction of digital health technologies offers a direct solution to many of these implementation barriers. A web-based WISN tool can automate the complex, multi-step calculations that currently require Excel expertise, provide immediate and visual feedback on staffing gaps, enable regular updates as workloads change, and present results in formats that are meaningful to managers without statistical backgrounds.

Research from Germany demonstrates that digital application of WISN not only produces accurate staffing calculations but also reveals opportunities for process optimization that can yield significant efficiency gains without additional resource expenditure (Thum, Wehner & Goetz, 2024).

The problem this project addresses is therefore both specific and significant: Nepalese hospitals lack accessible, practical tools to implement WHO WISN methodology for evidence-based nursing workforce planning. The gap between WISN's theoretical validity and its practical application at the facility level represents a missed opportunity for improved staffing management, better patient care, and more efficient use of Nepal's limited healthcare resources. Developing a user-friendly web-based WISN tool is a necessary and achievable step toward closing this gap.

Chapter 3: Objectives

3.1 General Objective

To develop a web-based prototype tool that applies WHO WISN methodology to calculate and visualize nursing staffing needs in hospital departments, demonstrating the practical application of health informatics in workforce planning for Nepalese healthcare settings.

3.2 Specific Objectives

1. To design a user-friendly web interface that simplifies WISN data input for hospital managers with limited technical background, incorporating clear guidance, validation, and step-by-step workflow design.
2. To develop automated calculation logic that converts manual WISN formulas into digital processes, including Available Working Time (AWT), Standard Workload, Staff Requirements, and WISN Ratio calculations.
3. To create interactive dashboards that clearly visualize staffing requirements, gaps, and workload pressure indicators using color-coded systems and graphical representations suitable for non-technical users.
4. To test the system's accuracy by comparing digital calculation outputs with manual WISN calculations performed using WHO-standard methodology on simulated hospital data from typical Nepalese hospital departments.
5. To document the system design, development process, and validation results comprehensively to facilitate future development, scaling, and potential institutional adoption by Nepalese hospitals.

Chapter 4: Literature Review

4.1 Health Workforce Crisis in Nepal

Nepal faces a critical health workforce shortage that has been documented extensively in national and international health literature. The nursing profession, which represents the largest cadre of clinical health workers in the country, is particularly affected by both absolute shortfalls in trained personnel and severe maldistribution between urban and rural settings. The nursing workforce is concentrated in urban centers, leaving rural health facilities severely understaffed and unable to deliver essential health services to the majority of Nepal's population, which remains predominantly rural.

The implications of this shortage extend beyond mere numbers. Research on nurses' workload and patient safety in Nepalese hospitals demonstrates that excessive patient-to-nurse ratios compromise the quality and safety of care delivered, contribute to high rates of nurse burnout and absenteeism, and increase turnover, further exacerbating the shortage (Shrestha & Kadel, 2019). An additional dimension of the workforce problem, documented in international research with potential applicability to Nepal, is the phenomenon of professional nurses being compelled to perform non-clinical tasks due to inadequate support staff. This task displacement effectively reduces the clinical capacity of professionally trained nurses, creating apparent nursing shortages that may in reality be support staff shortages (Ravhengani & Mtshali, 2017).

Nepal's Ministry of Health and Population has recognized these challenges in successive health sector plans and human resources for health strategies. However, systematic implementation of evidence-based workforce planning tools at the facility level has remained limited. The absence of reliable, real-time data on health worker activities and workloads makes it difficult to translate national-level policies into appropriate facility-level staffing decisions. This creates a persistent disconnect between policy intent and operational reality in Nepalese hospitals.

4.2 The WHO WISN Methodology

The Workload Indicators of Staffing Need (WISN) methodology was developed by the World Health Organization to provide a systematic, evidence-based approach to calculating health workforce requirements. Unlike traditional staffing norms based on population ratios or fixed establishments, WISN grounds its calculations in the actual work performed by health workers, making it responsive to the specific service profile and workload intensity of any given health facility or department (WHO, 2010, 2023).

The WISN methodology operates through an eight-step process. First, the health worker cadre and facility type are identified. Second, available working time is calculated by determining total working days per year and subtracting all non-working days including public holidays, annual leave, sick leave, and training days. Third, the workload components are defined to capture all the main activities performed by the cadre. Fourth, activity standards are established, representing the time in minutes or hours required to perform each workload component according to professional standards. Fifth, standard workloads are calculated by dividing available working time by the activity standard for each component. Sixth, allowance factors are calculated to account for additional activities and support functions. Seventh, the required number of staff is determined by dividing actual annual workload by standard workload and adding allowance factors. Eighth, the WISN ratio is calculated as current staff divided by required staff, with a ratio of 1.0 indicating balanced staffing, below 1.0 indicating shortage, and above 1.0 indicating surplus (WHO, 2010).

Studies across multiple health system contexts confirm that WISN serves as an effective and credible basis for evidence-based decision-making in human resources management. Its transparency and grounding in actual service data make it broadly applicable across primary, secondary, and tertiary levels of care. Research from Germany demonstrated that applying WISN to a hospital neurology therapy team provided an accurate quantification of required personnel resources while

simultaneously identifying inefficiencies in how existing staff time was being used (Thum, Wehner & Goetz, 2024).

4.3 Global Implementation of WISN

WISN has been implemented or piloted in numerous countries across multiple world regions, providing a valuable evidence base regarding both the methodology's potential and the practical challenges of its adoption. A comprehensive multi-country analysis examined WISN implementation experiences in India, South Africa, and Peru, identifying common implementation barriers and success factors (Mabunda et al., 2021).

In India, a pilot implementation of WISN in three district health facilities ultimately failed to produce sustainable results due to inadequate stakeholder consultation during the design phase, significant gaps in existing health information systems that made obtaining reliable workload data difficult, insufficient technical support for facility managers tasked with performing calculations, and challenges in defining appropriate workload components for the local service context. These challenges highlight the importance of contextual adaptation and capacity building in any WISN implementation effort.

South Africa's experience with WISN offers instructive lessons at the system scale. Following adoption of WISN for primary health care workforce planning between 2012 and 2015, the calculated staffing requirements were found to be substantially higher than current funded positions in all provinces, with one province reporting a gap requiring over five billion South African Rand to address. The affordability disconnect led to WISN being abandoned from the 2030 Human Resources for Health Strategy, underscoring that WISN results must be paired with realistic resource planning and broader stakeholder consultation including finance and treasury ministries (Mabunda et al., 2021). A qualitative study of WISN implementation in KwaZulu-Natal province found, however, that WISN had generated important insights about nursing workforce

utilization, including the counterintuitive finding that apparent nurse shortages were in some cases driven by nurses performing non-clinical activities due to support staff shortfalls (Ravhengani & Mtshali, 2017).

Peru's experience illustrates how WISN's data requirements can drive system improvement even when full implementation is not achieved. Faced with insufficient data for standard WISN calculations, Peru developed a context-specific alternative methodology using its National Register of Health Personnel (INFORHUS), demonstrating how engagement with WISN can catalyze broader health information system development (Mabunda et al., 2021).

A study applying WISN to maternal and child health services in Ganjam District, India found significant staffing shortfalls across multiple cadres, with particularly acute shortages among doctors, laboratory staff, and nurse midwives. The study generated locally validated activity standards and time measurements that were subsequently applicable to other Indian districts, demonstrating WISN's potential to produce contextually relevant benchmarks when properly implemented (Hagopian et al., 2012).

4.4 Digital Health and WISN Implementation

The integration of digital health technologies with workforce planning methodologies represents a significant and underexplored area of health informatics research. Digital tools have the capacity to address several fundamental limitations of manual WISN implementation: they can automate complex multi-step calculations, eliminate human arithmetic errors, enable regular updates as workloads change, present results through visual interfaces that are more accessible to non-technical users, and facilitate integration with existing hospital information systems.

The literature on digital WISN tools remains limited, reflecting the early stage of this application area. However, evidence from related fields of health informatics confirms that digitizing evidence-based clinical and managerial tools can substantially increase

their adoption and utilization. Barriers to adoption identified across health informatics literature include tools that are too technically complex for target users, inadequate training and support, poor interface design, and misalignment between tool outputs and the practical decisions managers need to make. Successful digital health tools are designed with end-user needs as the central design principle, tested iteratively with representative users, and accompanied by adequate training support (Ravhengani & Mtshali, 2017).

The German experience reported by Thum, Wehner and Goetz (2024) provides the most directly relevant evidence of digital WISN application. Their implementation in a hospital neurology therapy department demonstrated that systematic workload analysis using WISN methodology could identify both accurate staffing requirements and specific process inefficiencies amenable to optimization. The study found that therapists were spending 5.5 hours per week on support activities including documentation and scheduling, time that could potentially be redirected to direct patient care through process improvements or task delegation. This example illustrates how a well-implemented WISN analysis, whether manual or digital, can yield insights that go beyond simple headcount calculations to inform broader workforce management strategies.

4.5 Research Gap

While WISN methodology has been validated across diverse international contexts and its effectiveness for evidence-based workforce planning is well-established, a significant research gap exists regarding digital implementations suitable for resource-constrained settings like Nepal. Most existing WISN studies report manual implementations relying on Excel spreadsheets or paper-based calculations. The limited digital WISN tools that have been developed are typically created for high-resource environments with reliable internet connectivity, advanced technical infrastructure, and users with high levels of digital literacy.

There is a clear and unaddressed need for simple, browser-based WISN tools that can function with basic internet connectivity, require no software installation or technical expertise to operate, provide results in formats meaningful to hospital managers with limited statistical backgrounds, and can be adapted to the specific service context and workload patterns of Nepalese hospitals. This project addresses that gap directly, making both a practical contribution to Nepalese hospital management and an academic contribution to the emerging field of digital WISN implementation in low-resource settings.

Chapter 5: Methodology

5.1 Study Design and Approach

This project employs an applied health informatics research design focused on prototype system development. The methodology integrates software engineering principles with health system requirements analysis to create a functional, validated web-based tool. The development process follows an iterative approach with continuous testing and refinement cycles, ensuring that the final product meets both technical standards and the practical usability requirements of its intended end-users: hospital administrators and nurse managers in Nepalese healthcare settings.

The research design is pragmatic in orientation, prioritizing the development of a working tool that demonstrates the practical application of health informatics principles to a real healthcare management challenge. The project does not involve data collection from human subjects, clinical trials, or manipulation of patient information. Instead, it applies established WHO methodology to simulated but realistic hospital scenarios, producing a validated prototype whose accuracy can be assessed through comparison with manual calculation standards.

5.2 System Development Steps

5.2.1 Requirements Analysis

The first phase of system development involves comprehensive requirements analysis to define what the tool must do, who will use it, and under what conditions it will operate. This analysis will be grounded in a systematic review of WHO WISN manuals (2010, 2023), published implementation studies, and consultation with nursing faculty members and hospital administrators familiar with Nepalese hospital management contexts.

Functional requirements will specify the calculations the system must perform, the data inputs required, and the outputs and visualizations to be produced. Non-functional requirements will address performance, usability, accessibility, and browser compatibility. User requirements will define the characteristics of typical users, their technical literacy levels, and the specific management decisions the tool is expected to support. This multi-dimensional requirements analysis will form the basis for all subsequent design and development decisions.

5.2.2 System Architecture Design

The system architecture will be designed to prioritize simplicity, accessibility, and compatibility with the limited technical infrastructure typical of Nepalese hospital settings. The web application will be entirely browser-based, requiring no software installation, and will be accessible from any device with a standard internet browser and internet connectivity. This design principle ensures that the tool can be used on existing hospital computers without requiring IT system changes or specialized hardware.

The technical implementation will follow a standard web application architecture comprising three integrated components. The front-end user interface will be developed using HTML, CSS, and JavaScript to create an intuitive, responsive interface accessible on both desktop computers and tablet devices. The calculation engine, implemented in JavaScript, will contain all WISN formulas and calculation logic, converting data entered by users into staffing requirement outputs in real time. The data visualization layer will use established charting libraries, specifically Chart.js or similar open-source alternatives, to generate interactive graphs, color-coded indicators, and dashboard displays that make staffing information immediately interpretable.

5.2.3 Implementation of WISN Calculation Logic

The core of the system will be the precise digital implementation of WHO WISN calculation formulas. Each formula will be translated directly from the WHO User's Manual (2010, 2023) into JavaScript functions, ensuring fidelity to the established methodology. The following calculations will be automated:

Available Working Time (AWT) Calculation: AWT represents the actual productive working hours available per health worker per year. The system will calculate AWT by determining total calendar working days, then subtracting all non-working days including public holidays, annual leave entitlement, sick leave allowance, and training or continuing education days. The resulting net working days are multiplied by the standard working hours per day to produce AWT in hours. The formula is: $AWT = (Total\ Working\ Days - Non-Working\ Days) \times Working\ Hours\ Per\ Day$.

Standard Workload Calculation: For each health service activity performed by nursing staff, the standard workload represents the maximum number of times that activity can be performed by one staff member in one year given the available working time. This is calculated as: $Standard\ Workload = AWT / Activity\ Standard\ Time$, where the activity standard time is expressed in hours per activity. Higher standard workloads indicate that a staff member can perform the activity more times per year, typically because the activity requires relatively less time.

Staff Requirements Calculation: The required number of staff for each workload component is calculated by dividing the annual volume of that activity by the standard workload. This produces the number of full-time equivalent staff positions needed solely to perform that activity. When requirements are summed across all workload components and adjusted by the allowance factor, the result represents the total staff required to deliver all services in a department: $Required\ Staff = Annual\ Workload / Standard\ Workload$.

WISN Ratio Calculation: The WISN ratio provides the key summary indicator of staffing adequacy. It is calculated by dividing current staff by required staff: $WISN$

Ratio = Current Staff / Required Staff. A ratio of exactly 1.0 indicates that current staffing precisely meets calculated requirements. Ratios below 1.0 indicate staffing shortfalls, with the degree of shortage proportional to the distance below 1.0. Ratios above 1.0 indicate staffing surplus. The system will automatically interpret and categorize WISN ratios using color-coded indicators, making results immediately understandable to non-technical users.

5.2.4 User Interface Design

The user interface will be designed with a hospital manager who has limited technical background as the central design persona. Every interface decision will prioritize clarity, simplicity, and guided workflow over technical sophistication. The interface will guide users through the WISN process step-by-step, presenting one logical section at a time rather than displaying all data fields simultaneously, which can be overwhelming.

Key design features will include clearly labeled data entry forms with helpful contextual tooltips explaining each field, immediate data validation that alerts users to missing or implausible entries before calculations proceed, a step-by-step workflow that mirrors the logical sequence of WISN calculation, visual dashboards with color-coded indicators using a standard green-yellow-red system for adequate, borderline, and critical staffing situations respectively, and the ability to generate printable summary reports suitable for presentation to hospital administration boards.

5.3 Data Collection and Testing Strategy

Since this is a prototype development project focused on demonstrating health informatics principles rather than conducting clinical research, no real patient data or personally identifiable staff information will be collected or used. All system testing will use simulated data designed to represent realistic Nepalese hospital scenarios. This approach is standard practice in health informatics prototype development and allows

thorough system validation without creating ethical concerns about patient privacy or data security.

5.3.1 Simulated Data Sources

The simulated data used for system testing and validation will be derived from multiple authoritative sources to ensure it represents realistic hospital conditions:

- **WHO WISN Manual Examples:** The WHO WISN User's Manuals (2010, 2023) contain detailed case study examples from multiple countries, including specific activity standards, workload volumes, and staffing calculations. These published examples provide validated reference data that will be adapted to reflect Nepalese hospital contexts, providing a solid empirical foundation for simulated test scenarios.
- **Published Research Data:** Academic studies implementing WISN in comparable low- and middle-income country contexts, including South Africa, India, Uganda, and Bangladesh, provide validated data on nursing activity types, time standards, and typical workload patterns across different hospital departments. Data from these studies will be referenced and adapted for simulated scenarios representing typical Nepalese hospital departments.
- **Ministry of Health Guidelines:** Nepal's Ministry of Health and Population has published general staffing norms, service delivery standards, and human resources for health policy documents that provide baseline information about expected staffing levels, service volumes, and nurse-to-patient ratios in Nepalese health facilities. These national standards will inform the simulated scenarios to ensure they are consistent with the national health policy context.
- **Expert Consultation:** Nursing faculty members with experience in hospital management and workforce planning, as well as hospital administrators familiar with day-to-day Nepalese hospital operations, will be consulted to

validate that simulated scenarios are realistic. This expert validation ensures that the activity standards, time assumptions, and workload volumes used in testing accurately reflect what nurses actually do in Nepalese hospitals.

5.3.2 Typical Hospital Scenarios for Testing

The system will be tested using simulated data from four typical Nepalese hospital departments, chosen to represent the diversity of nursing workloads across different care settings:

Intensive Care Unit (ICU): High-intensity nursing care including continuous patient monitoring, ventilator management, medication administration, vital signs recording every hour, intravenous line care, and family communication. Simulated data will reflect typical Nepalese tertiary hospital ICU patient loads and standard nursing time measurements from international literature adapted for local conditions.

Emergency Department: Fast-paced, high-volume environment with activities including patient triage, wound care, medication administration, patient stabilization, intravenous line insertion, and documentation. Simulated data will reflect typical emergency visit volumes, acuity mix, and nursing time standards appropriate for a Nepalese district or regional hospital.

General Medical Ward: Inpatient care including patient assessments, medication rounds, wound dressing, vital signs monitoring, patient education, discharge planning, and documentation. Simulated bed occupancy rates and nursing activity volumes will be based on typical medical ward conditions in Nepalese public hospitals.

Surgical Ward: Post-operative care including surgical wound dressing changes, pain assessment and management, post-operative vital signs monitoring, patient mobilization assistance, and patient discharge preparation. Simulated data will represent typical surgical case volumes and recovery nursing requirements for Nepalese hospital settings.

5.3.3 System Validation Approach

A systematic, multi-stage validation process will be implemented to ensure the digital tool produces calculations that are both accurate and meaningful:

- **Manual Calculation Comparison:** For each test scenario, WISN calculations will first be performed manually using Microsoft Excel spreadsheets, strictly following WHO methodology. The digital tool's outputs for identical inputs will then be compared against these manual calculations. Any discrepancies will be investigated to determine whether they represent formula errors in the digital tool, data entry differences, or rounding differences, and all discrepancies will be resolved before the tool is considered validated.
- **Formula Verification:** Each calculation step implemented in the digital tool will be traced back to the specific formula published in the WHO WISN User's Manual (2010, 2023). A verification table documenting each formula's source, its digital implementation, and test results will be compiled as part of the validation documentation.
- **Expert Review:** Nursing faculty members with experience in health workforce planning will review a sample of calculations generated by the tool, confirming that outputs are numerically reasonable and that interpretations of WISN ratios are appropriate for the Nepalese health system context.
- **Usability Testing:** A small group of hospital administrators and nurse managers will be invited to interact with the tool prototype, providing structured feedback on ease of use, clarity of instructions, comprehensibility of visual outputs, and the relevance of generated reports to real-world staffing decisions. This feedback will guide interface refinements before finalization.

5.4 Technical Tools and Software

The project will utilize exclusively free and open-source technologies to ensure that the final product can be replicated, maintained, and extended without requiring expensive software licenses. Core technologies will include HTML5, CSS3, and JavaScript for front-end interface development; Chart.js or D3.js for interactive data visualization; Node.js for any required server-side functionality; and GitHub for version control, code documentation, and project management. Microsoft Excel will be used exclusively for manual calculation validation purposes, not as part of the digital tool itself.

5.5 Ethical Considerations

This project involves no direct interaction with patients, no collection of personally identifiable information from health workers, and no use of sensitive clinical data. All data used for system development, testing, and demonstration purposes will be simulated or derived from publicly available published sources. Because no human subjects are involved and no identifiable data is collected, formal ethical clearance from an institutional review board is not required for this prototype development project. However, the project will adhere to all relevant ethical research principles including complete transparency about data sources and limitations, appropriate attribution of all referenced materials, acknowledgment of contributions from faculty advisors and expert consultants, and responsible representation of the tool's capabilities and limitations in all documentation and presentations.

Chapter 6: Expected Outcomes

This project will produce a set of concrete, interrelated deliverables that together demonstrate the practical application of health informatics in addressing a real healthcare management challenge in Nepal.

6.1 Functional Web-Based Prototype

The primary deliverable will be a fully working web application accessible through standard internet browsers without requiring software installation. The prototype will include a user-friendly data entry system for hospital department information, an automated WISN calculation engine that processes inputs and produces outputs in real time, interactive dashboards displaying staffing requirements by department with visual gap indicators, WISN ratio calculations presented with clear interpretive guidance, and a printable summary report function for administrative use. The prototype will be tested and validated for calculation accuracy before final submission.

6.2 System Documentation

Comprehensive documentation will be prepared alongside the working prototype, including a system architecture document describing all technical design decisions, a user manual with step-by-step instructions designed for hospital managers with limited technical backgrounds, a technical reference document for IT staff covering system maintenance and potential customization, and a validation report presenting the comparison between digital and manual WISN calculations across all test scenarios.

6.3 Academic Contribution

Beyond its practical deliverables, the project will contribute to the emerging body of literature on digital WISN implementation in resource-constrained settings. The project

documentation, methodology, and validation results will be prepared in a format suitable for dissemination to academic audiences, providing a model for future digital WISN development efforts in similar settings.

Chapter 7: Work Plan and Timeline

The project will be completed over a twelve-week period organized into five sequential phases, each building on the outputs of the previous phase. The timeline has been designed to be realistic given the scope of prototype development while ensuring adequate time for validation and documentation.

Phase 1: Literature Review and Requirements Analysis (Weeks 1-2)

- Conduct systematic review of WHO WISN manuals (2010, 2023) and selected published implementation studies
- Consult with nursing faculty members and hospital administrators to understand practical requirements
- Compile and organize simulated test data from published sources
- Develop complete functional and non-functional requirements specifications
- Prepare manual WISN calculations for all planned test scenarios using Excel

Phase 2: System Design (Weeks 3-4)

- Design overall system architecture and data flow diagrams
- Create user interface wireframes and mockups for all key screens
- Plan database structure for storing department and calculation data
- Design validation test cases with expected outputs derived from manual calculations

- Review and finalize design specifications with faculty advisors

Phase 3: Development (Weeks 5-8)

- Develop front-end user interface following approved wireframes
- Implement WISN calculation engine with all required formulas
- Integrate data visualization dashboards and WISN ratio indicators
- Conduct iterative unit testing of each calculation component
- Integrate all system components and conduct integration testing

Phase 4: Validation and Testing (Weeks 9-10)

- Compare digital calculation outputs against manual Excel calculations for all test scenarios
- Document discrepancies and implement corrections
- Conduct usability testing with potential end-users
- Gather faculty advisor feedback and implement recommended refinements
- Prepare final validation report

Phase 5: Documentation and Finalization (Weeks 11-12)

- Complete all system documentation including user and technical manuals
- Finalize validation report and academic write-up
- Prepare project demonstration materials and presentation

- Submit all deliverables for faculty review

Chapter 8: Significance of the Project

This project addresses a critical gap in Nepalese hospital management by improving the usability and accessibility of an internationally validated workforce planning methodology. Its significance extends across multiple stakeholder groups and dimensions of the health system.

8.1 For Hospital Management

The web-based tool will enable hospital administrators and nurse managers to move beyond intuition-based staffing decisions toward evidence-based management. With an accessible digital WISN tool, managers will be able to identify departments experiencing excessive workload pressure before staffing crises develop, allocate nursing resources more equitably across departments, justify staffing requests and budget submissions to hospital boards with concrete, internationally validated data, and monitor staffing adequacy regularly as patient volumes and service mixes change over time.

8.2 For Nursing Staff

Improved staffing analysis directly benefits the nurses whose working conditions and professional satisfaction depend on appropriate staffing levels. Evidence-based identification of understaffed departments provides objective justification for staffing improvements that management intuition alone may not support. More balanced workload distribution reduces the burnout and turnover that perpetuate staffing shortfalls, and adequate staffing creates the conditions necessary for delivering safe, quality patient care.

8.3 For Nepal's Health System

At the health system level, this project contributes to strengthening Nepal's health workforce planning capacity by making an internationally validated methodology practically accessible at the facility level. The project aligns with WHO recommendations for evidence-based workforce planning and supports Nepal's broader digital health transformation agenda by demonstrating how health informatics can be applied to address workforce management challenges. A validated, open-source digital WISN prototype could be replicated, adapted, and scaled across Nepal's hospital system if further institutional support is secured.

8.4 For Health Informatics Education

The project provides an applied educational example of how health informatics principles can be brought to bear on real healthcare management challenges. It demonstrates the complete lifecycle of health informatics application development, from requirements analysis through implementation and validation, and illustrates the practical challenges and solutions involved in translating evidence-based clinical and management methodologies into accessible digital tools.

Chapter 9: Limitations

This project acknowledges several limitations that define its scope and should be understood by any audience evaluating its outputs or considering future development:

Prototype Development Only: This project produces a functional prototype for academic demonstration purposes. It does not include full-scale system deployment in actual hospital settings. Real-world implementation would require additional phases including security hardening, privacy impact assessment, integration with existing hospital information systems, comprehensive user training programs, and sustained technical support infrastructure.

Simulated Data: All system testing uses simulated data derived from published literature and expert consultation rather than actual hospital operational records. While simulated scenarios are carefully designed to be realistic and contextually appropriate, they cannot fully replicate the complexity, variability, and specificity of actual hospital data. Validation with real facility data would be a necessary step before any operational deployment.

Nursing Staff Focus: This project specifically addresses nursing workforce planning and does not extend to other health worker cadres including physicians, laboratory technicians, pharmacists, or administrative and support staff. While the methodology is in principle applicable to any cadre, the tool's design and validation focus exclusively on nursing.

Selected Departments: The prototype demonstrates calculations for four common hospital departments rather than all possible service areas within a complete hospital. The selection prioritizes representative diversity over comprehensiveness, with the expectation that the methodology demonstrated can be extended to additional departments in future development.

Quantitative Focus: WISN methodology is inherently quantitative, focusing on workload volumes and time standards while not capturing qualitative dimensions of nursing care such as patient acuity variation, emotional support requirements, care coordination complexity, or the unique demands of managing complex, multi-morbid patients. These qualitative factors are important complements to quantitative staffing analysis and should be considered alongside WISN outputs in actual staffing decisions.

Chapter 10: Conclusion

This proposal presents a compelling case for developing a web-based nursing staffing tool that implements WHO WISN methodology in an accessible, user-friendly digital format designed for the specific context of Nepalese hospitals. By combining the scientific rigor of an internationally validated workforce planning methodology with the practical accessibility of modern health informatics technology, this project aims to bridge a significant gap between theoretical knowledge and operational hospital management practice.

The problem the project addresses is both real and consequential. Nepalese hospitals face nursing workforce challenges that cannot be effectively addressed through intuition-based staffing decisions or periodic manual WISN calculations performed as isolated exercises. What is needed is a practical, accessible tool that enables hospital managers to perform evidence-based staffing analysis as part of their routine management practice. A digital WISN tool designed with the specific needs and constraints of Nepalese hospital managers in mind is a necessary and achievable step toward this goal.

The methodology for prototype development is sound, drawing on WHO-established formulas, realistic simulated data derived from authoritative published sources, and a systematic validation process that compares digital outputs against manual calculations. The use of simulated data is not a limitation to apologize for but a deliberate and appropriate choice for a prototype development project that must demonstrate technical accuracy without creating ethical concerns about patient privacy or data security.

The project's significance extends across multiple dimensions and stakeholder groups: hospital managers gain accessible tools for evidence-based decision-making, nursing staff benefit from the potential for more equitable and adequate staffing, Nepal's health system gains a practical demonstration of digital health application to workforce

management, and the health informatics field gains an applied example of digital WISN implementation in a resource-constrained setting.

While the project acknowledges its limitations as an academic prototype, it also provides a concrete foundation for future development. The working prototype, comprehensive documentation, and validated calculations produced by this project could form the basis for institutional adoption, scaling, and enhancement if adequate support is secured from hospitals, health authorities, or development partners interested in strengthening Nepal's health workforce planning capacity.

In conclusion, this project represents a meaningful and feasible contribution to improving nursing workforce planning in Nepal through the thoughtful application of health informatics principles to an internationally recognized methodology. It is grounded in evidence, responsive to real-world needs, technically achievable within the project timeframe, and positioned to generate lasting value beyond its immediate academic context.

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